

Tyler Nguyen

757-266-9690 | wgq4tr@virginia.edu | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Virginia

B.A. in Computer Science | **GPA: 3.9/4.0** | **SAT: 1570**

Charlottesville, VA

Expected May 2028

PROFESSIONAL EXPERIENCE

Software Engineer Intern | BNSF Railway

June 2026 - July 2026

- Migrated 60+ batch jobs across 3 applications from legacy Java on WebSphere to Spring Boot on OpenShift, contributing to a Kyndryl contract exit projected to save **\$55M**.
- Cut batch job runtime **60%** by batching database writes across 4 write paths instead of one round trip per row.
- Wrote **156 tests across 32 classes** and dry-run flags that verified migrated jobs matched legacy output.

Software Engineer Intern | Booz Allen Hamilton

April 2026 - May 2026

- Engineered a **multi-provider LLM gateway** in LiteLLM routing across Ollama, Gemini 2.5 Flash, and GPT-4o mini, with per-model rate and spend limits and Keycloak role-based access.
- Developed a **Strands agent** with MCP tools and a React chat UI, using Phoenix tracing for latency and spend.
- Implemented a **two-layer guardrail** pairing a local keyword filter with an LLM classifier for ambiguous prompts.

AI Consultant | MyAiPathways

January 2026 - April 2026

- Reduced manual applicant review by **75%** with a Python **resume screening pipeline** that extracts skills with spaCy, measures job fit with sentence-transformers, and grades candidates against an LLM rubric.
- Launched a full-stack **document automation platform** in vanilla JavaScript with a Node/SMTP service that drafts and emails offer letters and client contracts **90%** faster.

Machine Learning Developer | ML@UVA

October 2025 - January 2026

- Created a Flutter app for the UVA Math Tournament that runs a **TensorFlow Lite CNN** on handwritten answer sheets, cutting grading from **three minutes to under ten seconds** per sheet.
- Achieved **100%** accuracy on slanted bubble grids and **96%** on blank cells via OpenCV contour digit cropping.

Software Engineer Intern | TheCourseForum

September 2025 - December 2025

- Deployed a Django **content moderation pipeline** that scores reviews with Detoxify across six weighted toxicity signals, auto-hiding posts past a 0.75 threshold and flagging **5%** of 32,000 reviews.
- Standardized front-end colors into shared CSS variables and shipped **dark mode** sitewide for a **20k+ user base**.

TECHNICAL PROJECTS

AI Short-Form Video Generator | [GitHub](#)

July 2026 - July 2026

- Shipped a Python pipeline that uses an **LLM-as-judge** to score Reddit stories on 5 engagement dimensions and render the best as short-form videos in **90%** less time than manual editing, at under \$0.05 per batch.

Rally | [HooHacks 2026](#) | [Live Demo](#) | [GitHub](#)

March 2026 - March 2026

- Built a **passive nightlife-safety app** that fuses gait, distance from friends, and ambient sound into a 0–100 risk score that gates a **Claude tool-use agent** costing under \$0.001 per check, which escalates across four alert tiers.

LEADERSHIP

Teaching Assistant, CS 2130 | University of Virginia

August 2026 - Present

- Lead weekly **lab sections** and **office hours** for a **300+ student** course, debugging C and circuit assignments.

Webmaster | FIND Inc District 7 | [Live Website](#)

August 2026 - Present

- Maintain and update the **district website** for Filipino student organizations across **6 Virginia universities**.

Technology Head | Project Code

April 2026 - Present

- Host **Git and project best-practice seminars**, LeetCode nights, and mock interviews for club members.

SKILLS

Languages: Python, Java, C, TypeScript, JavaScript, SQL, R, HTML/CSS

Frameworks/Libraries: React.js, Node.js, Express.js, Flask, Django, Spring Boot, OpenCV, LiteLLM, PyTorch

Developer Tools: Git, Docker, Kubernetes, OpenShift, Rancher, Jenkins, PostgreSQL, Keycloak, Linux